



JKBOSE 2026

CLASS 11TH

STRAIGHT LINES

MATHS

LECTURE-03

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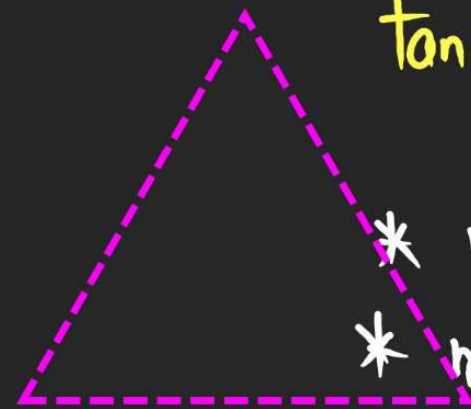
TOPICS TO BE COVERED

1

Exercise Discussion 9.1

2

Related Problems

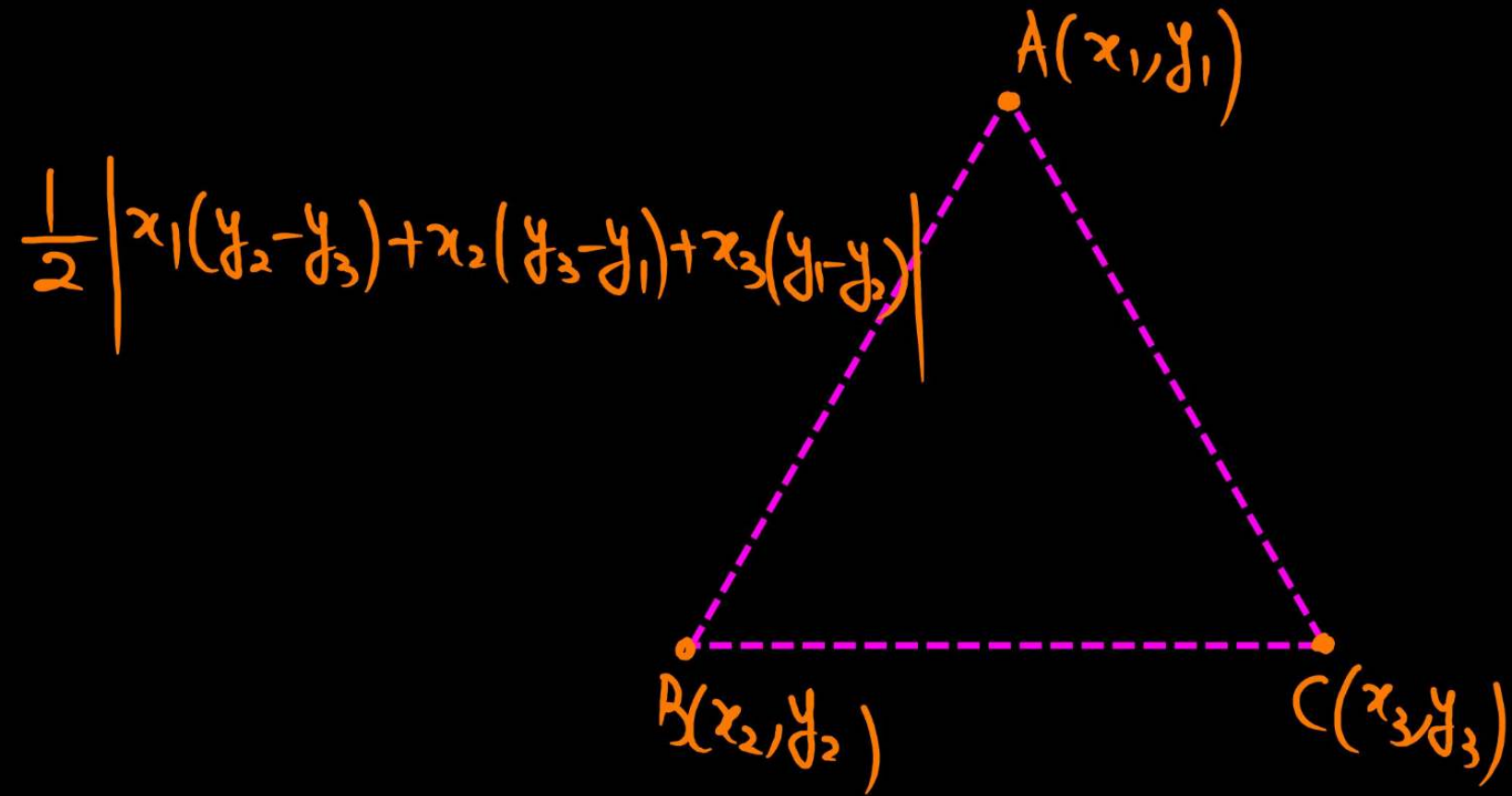


$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right| \quad \left| \frac{2m - m}{1 + 2m \times m} \right|$$

* $m_1 = m_2$ (Parallel lines)

* $m_1 m_2 = -1$ (Perpendicular lines)

$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1} = m.$$



EXERCISE – 9.1

QUESTION

Draw a quadrilateral in the Cartesian plane, whose vertices are $(-4, 5)$, $(0, 7)$, $(5, 5)$ and $(-4, -2)$. Also, find its area.

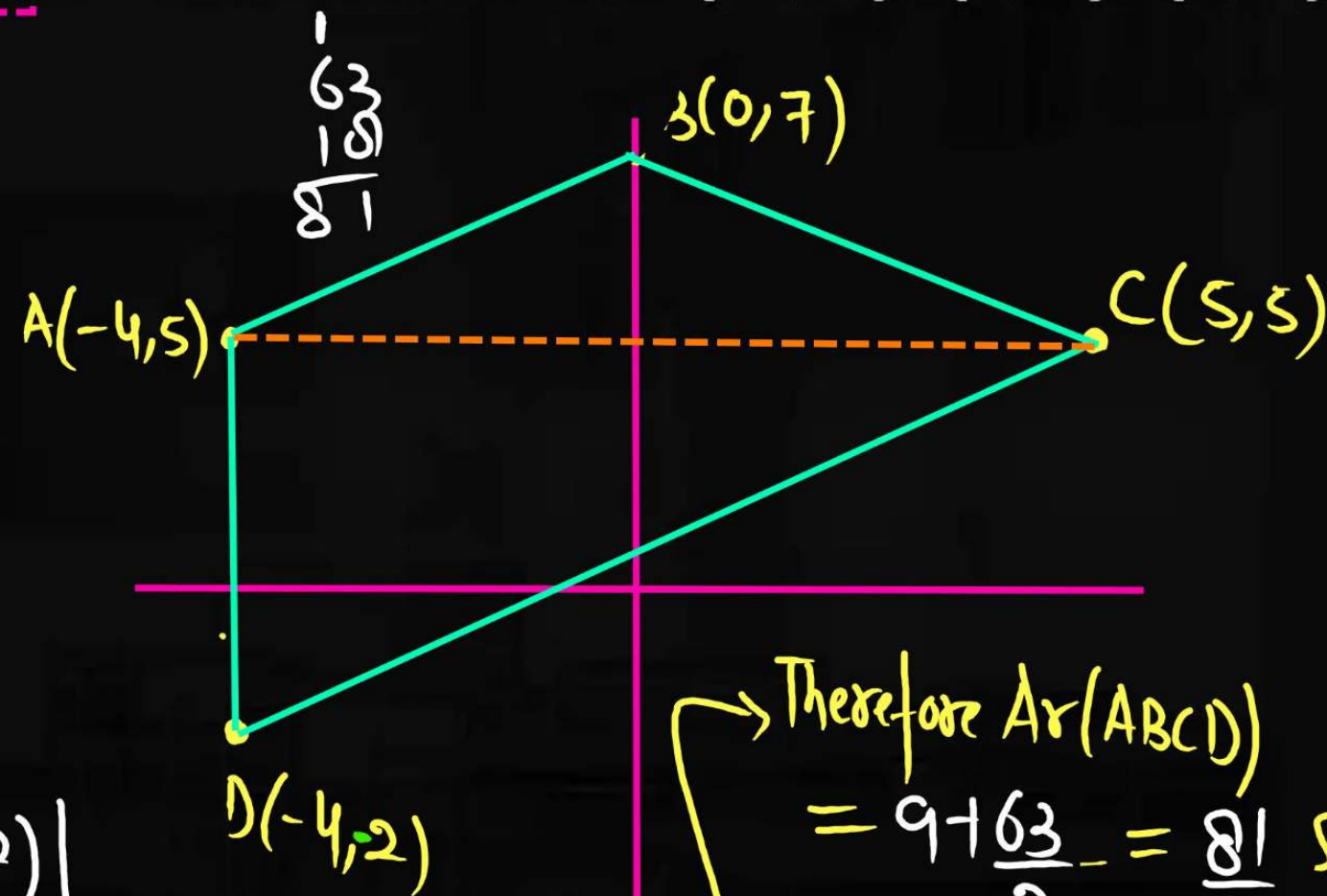
(Sol) Here $\text{Ar}(\triangle ABC) = \frac{1}{2} |-4(7-5) + 0(5-5) + 5(5-7)|$

$$\Rightarrow \text{Ar}(\triangle ABC) = \frac{1}{2} |-8 - 10| = \frac{1}{2} \times 18 = 9$$

$$\Rightarrow \text{Ar}(\triangle ABC) = 9 \text{ Sq. Units.}$$

Also $\text{Ar}(\triangle ADC) = \frac{1}{2} |-4(2-5) - 4(\cancel{5}-\cancel{5}) + 5(5+2)|$

$$\Rightarrow \text{Ar}(\triangle ADC) = \frac{1}{2} |20 + 35| = \frac{55}{2} \text{ Sq. Units.}$$



Therefore $\text{Ar}(ABCD) = 9 + \frac{55}{2} = \frac{69}{2} \text{ Sq. Unit.}$

QUESTION

The base of an equilateral triangle with side $2a$ lies along the y -axis such that the mid-point of the base is at the origin. Find vertices of the triangle.

(Sol) Clearly $\triangle AOB$ is a right Triangle,

So $AB^2 = OA^2 + OB^2$ (By Pythagoras theorem)

$$\Rightarrow OA^2 = AB^2 - OB^2$$

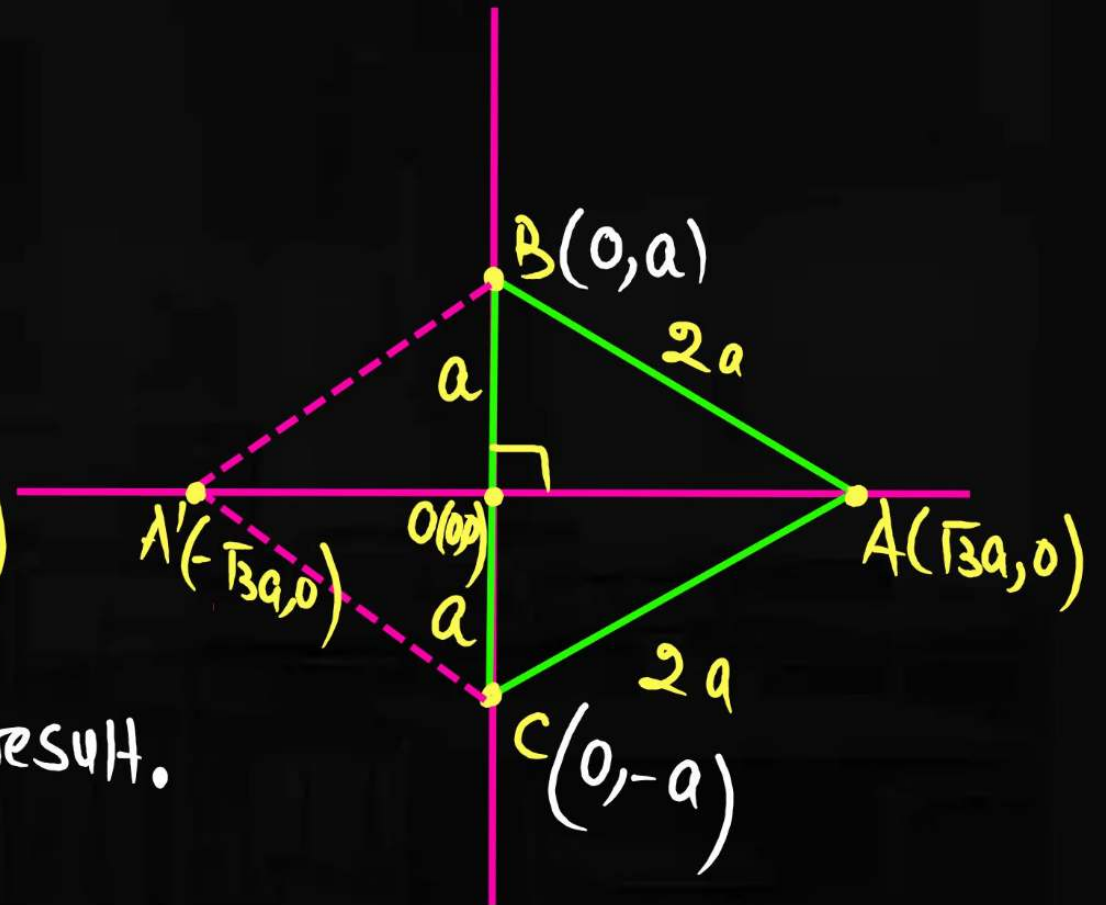
$$\Rightarrow OA^2 = 4a^2 - a^2$$

$$\Rightarrow OA^2 = 3a^2$$

$$\Rightarrow OA = \sqrt{3}a$$

Therefore the co-ordinates of the vertices are $A(\pm\sqrt{3}a, 0)$, $B(0, a)$ & $C(0, -a)$

Hence the result.



QUESTION

Find the distance between $P(x_1, y_1)$ and $Q(x_2, y_2)$ when:

- (i) PQ is parallel to the y-axis,
- (ii) PQ is parallel to the x-axis.

Homework

QUESTION



Find a point on the x -axis, which is equidistant from the points $(7, 6)$ and $(3, 4)$.

(Sol) let $P(a, 0)$ be the required point on the x -axis which is equidistant from $A(7, 6)$ and $B(3, 4)$. So $AP = BP$

$$\Rightarrow AP^2 = BP^2$$

$$\Rightarrow (a-7)^2 + (0-6)^2 = (a-3)^2 + (0-4)^2$$

$$\Rightarrow a^2 + 49 - 14a + 36 = a^2 + 9 - 6a + 16$$

$$\Rightarrow 85 - 14a = 25 - 6a$$

$$\Rightarrow -14a + 6a = 25 - 85$$

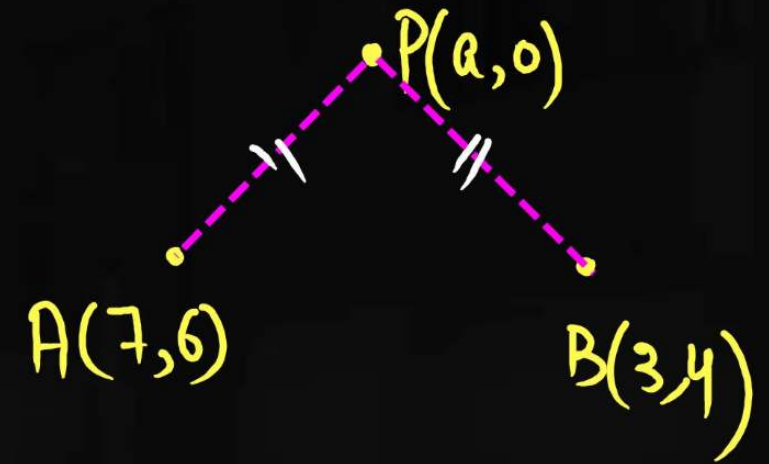
$$\Rightarrow -8a = -60$$

$$\Rightarrow a = \frac{60}{8}$$

$$\Rightarrow a = \frac{15}{2}$$

Therefore the required point $P(\frac{15}{2}, 0)$

Hence the result



QUESTION

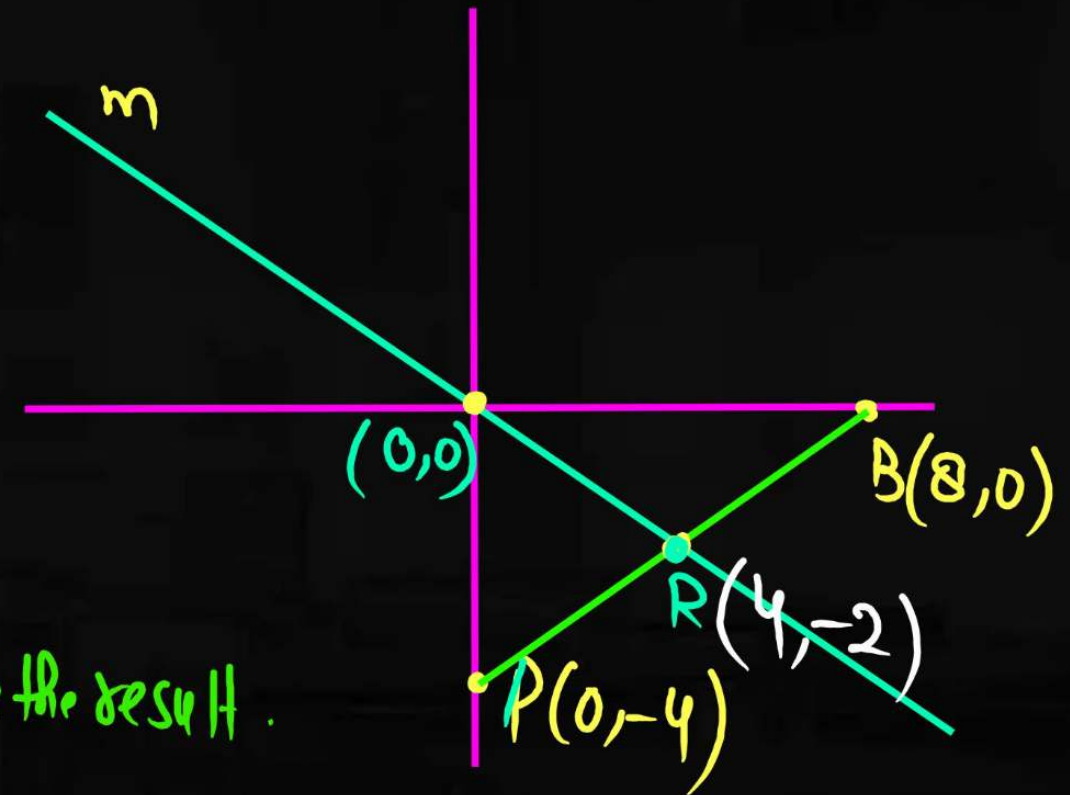
Find the slope of a line which passes through the origin and the mid-point of the line segment joining the points $P(0, -4)$ and $B(8, 0)$.

(Sol) By Mid-point formula,

$$R\left(\frac{0+8}{2}, \frac{0-4}{2}\right) = R(4, -2)$$

Therefore the required slope = $-\frac{2-0}{4-0} = -\frac{2}{4} = -\frac{1}{2}$.

Hence the result.



QUESTION

Without using the Pythagoras theorem, show that the points $(4, 4)$, $(3, 5)$ and $(-1, -1)$ are the vertices of a right angled triangle.

(Sol) let the given vertices be $A(4, 4)$, $B(3, 5)$ & $C(-1, -1)$. Then

$$m_{AB} = \frac{5-4}{3-4} \Rightarrow m_{AB} = -1$$

$$\text{Also } m_{BC} = \frac{-1-5}{-1-3} = -\frac{6}{-4} = \frac{3}{2}$$

$$m_{AC} = \frac{-1-4}{-1-4} \Rightarrow m_{AC} = 1$$

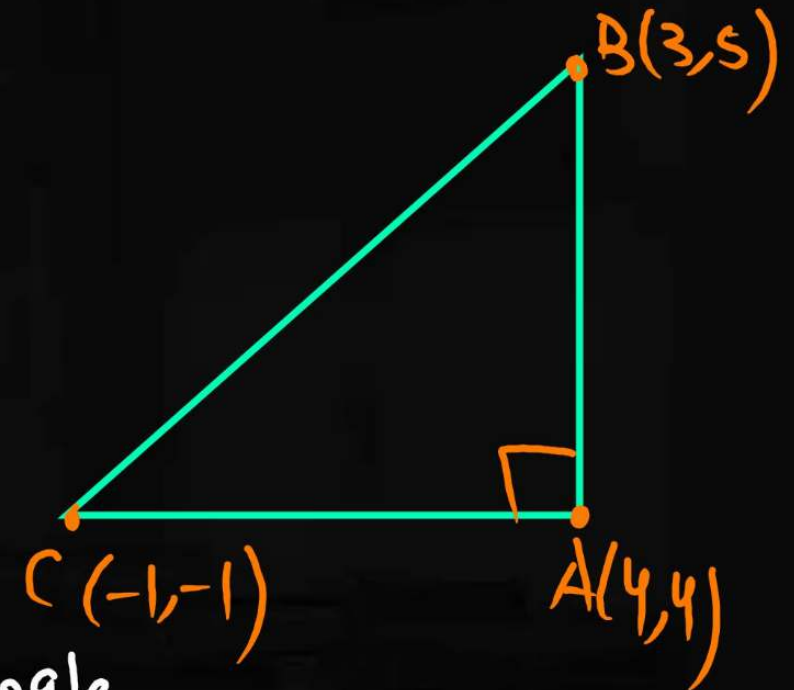
$$\text{Here, } m_{AB} \times m_{AC} = -1 \times 1$$

$$\Rightarrow m_{AB} \times m_{AC} = -1$$

$$\Rightarrow AB \perp AC$$

$\Rightarrow \triangle ABC$ is a right angled Triangle.

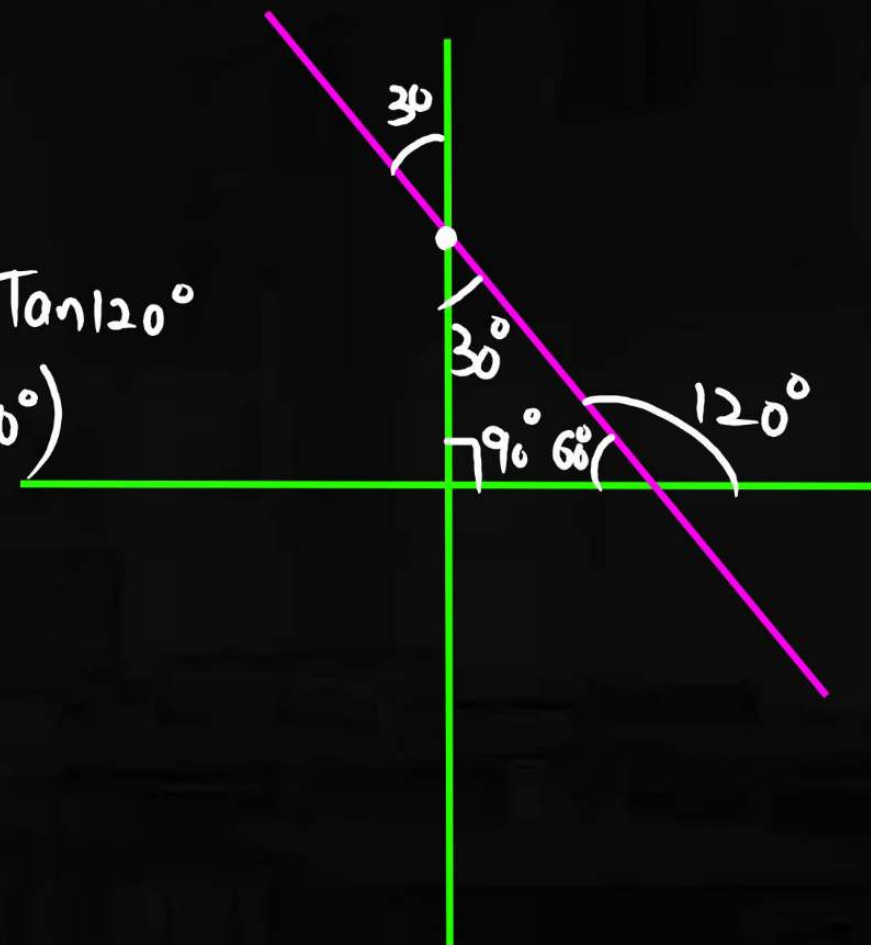
Hence proved.



QUESTION

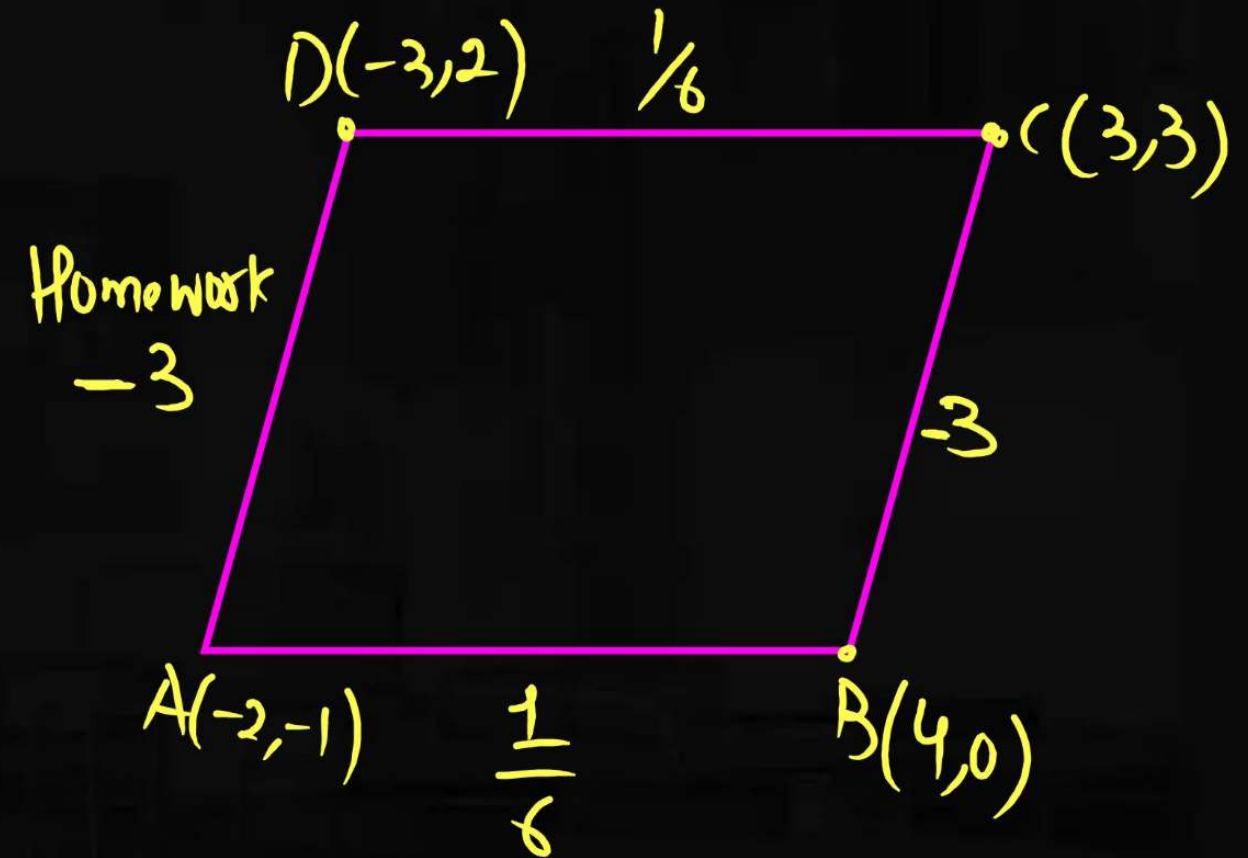
Find the slope of the line, which makes an angle of 30° with the positive direction of y-axis measured anticlockwise.

$$\begin{aligned}
 m &= \tan 120^\circ \\
 \Rightarrow m &= \tan(90 + 30^\circ) \\
 \Rightarrow m &= -\cot 30^\circ \\
 \Rightarrow m &= -\sqrt{3}
 \end{aligned}$$



QUESTION

Without using distance formula, show that points $(-2, -1)$, $(4, 0)$, $(3, 3)$ and $(-3, 2)$ are the vertices of a parallelogram.



QUESTION

Find the angle between the x -axis and the line joining the points $(3, -1)$ and $(4, -2)$.

H/w

QUESTION

The slope of a line is double of the slope of another line. If tangent of the angle between them is $1/3$ find the slopes of the lines.

(sol) let the given slopes be m & $2m$ such that

$$\tan \theta = \frac{1}{3}$$

$$\Rightarrow \left| \frac{2m - m}{1 + 2m \times m} \right| = \frac{1}{3}$$

$$\Rightarrow \left| \frac{m}{1 + 2m^2} \right| = \frac{1}{3}$$

$$\frac{m}{1 + 2m^2} = \pm \frac{1}{3}$$

Case 1st: $\frac{m}{1 + 2m^2} = \frac{1}{3}$

$$\Rightarrow 3m = 1 + 2m^2$$

$$\Rightarrow 2m^2 - 3m + 1 = 0$$

$$\Rightarrow 2m^2 - 2m - m + 1 = 0$$

$$\Rightarrow 2m(m-1) - 1(m-1) = 0$$

$$\Rightarrow (2m-1)(m-1) = 0$$

$$\Rightarrow m = \frac{1}{2}, 1 \text{ \& } 2m = 1, 2.$$

Therefore the required slopes are $\frac{1}{2}, 1, 1, 2$ in this case

Case 2nd: $\frac{m}{1 + 2m^2} = -\frac{1}{3}$

$$\Rightarrow 1 + 2m^2 = -3m$$

$$\Rightarrow 2m^2 + 3m + 1 = 0$$

$$\Rightarrow (2m+1)(m+1) = 0$$

$$\Rightarrow m = -\frac{1}{2}, -1 \text{ \& } 2m = -1, -2.$$

Therefore the req. slopes are $-\frac{1}{2}, -1, -1, -2$ in this case

QUESTION



A line passes through (x_1, y_1) and (h, k) . If slope of the line is m , show that

$$k - y_1 = m (h - x_1).$$

H/w



$$m = \frac{k - y_1}{h - x_1}$$

$$\Rightarrow k - y_1 = m(h - x_1)$$

QUESTION(NDA-2021)

A parallelogram has three consecutive vertices $(-3, 4)$, $(0, -4)$ and $(5, 2)$. The fourth Vertex is :

1 ✓ $(2, 10)$

2 ✗ $(2, 9)$

3 ✗ $(3, 9)$

4 ✗ $(4, 10)$

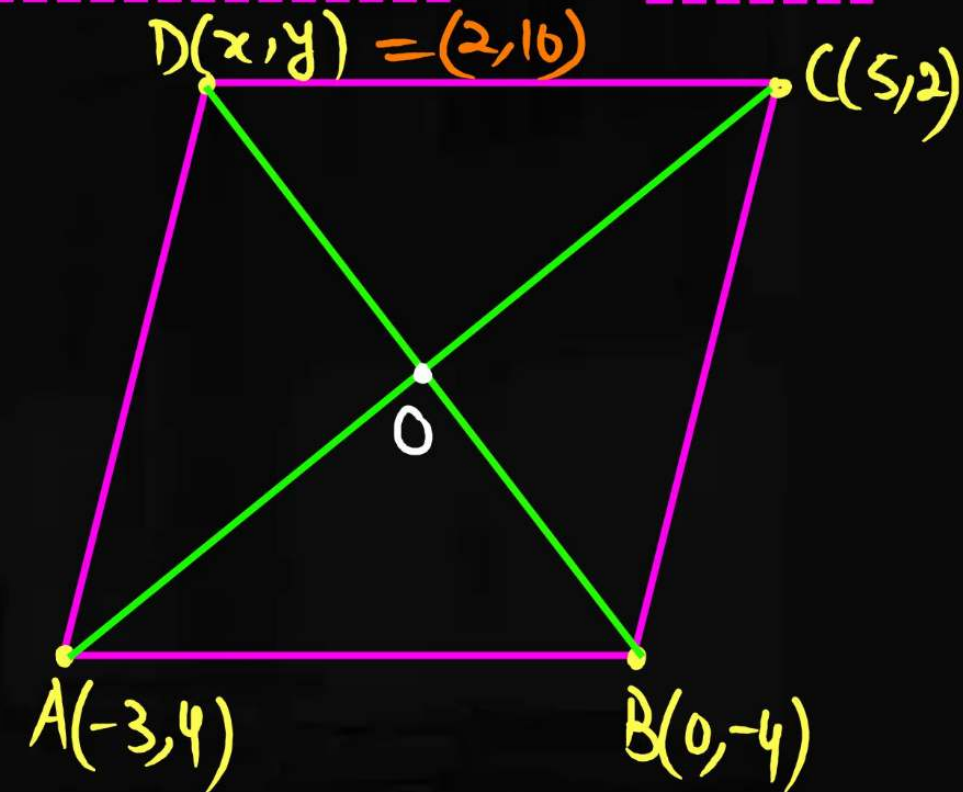
(Sol) Since we know that the diagonals of a llgm bisects each other, then O is the mid-point of AC & BD

$$\Rightarrow O\left(\frac{5-3}{2}, \frac{2+4}{2}\right) = O\left(\frac{x}{2}, \frac{y-4}{2}\right)$$

$$\Rightarrow O(1, 3) = O\left(\frac{x}{2}, \frac{y-4}{2}\right)$$

$$\text{So, } \frac{x}{2} = 1 \text{ \& } \frac{y-4}{2} = 3$$

$$\Rightarrow x = 2, y = 10.$$

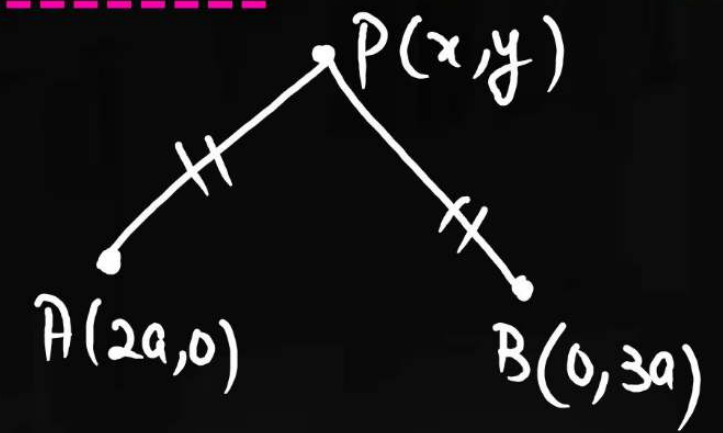


QUESTION(NDA-2022)

If the point (x, y) is equidistant from the points $(2a, 0)$ and $(0, 3a)$, where $a > 0$, then which one of the following is correct?

- 1 ☒ $2x - 3y = 0$
- 2 ☒ $3x - 2y = 0$
- 3 ☒ $4x - 6y + 5a = 0$
- 4 ☒ $4x - 6y - 5a = 0$

(Sol) Here $PA = PB$
 $\Rightarrow PA^2 = PB^2$
 $\Rightarrow (x-2a)^2 + y^2 = x^2 + (y-3a)^2$
 $\Rightarrow \cancel{x^2} + 4a^2 - 4ax + \cancel{y^2} = \cancel{x^2} + \cancel{y^2} + 9a^2 - 6ay$
 $\Rightarrow 4ax - 6ay + 5a^2 = 0$
 $\Rightarrow 4x - 6y + 5a = 0$



QUESTION

A light ray emerging from the point source placed at $P(2, 3)$ is reflected at a point Q on the y -axis. It then passes through the point $R(5, 10)$. The coordinates of Q are :

- 1 $(0, 3)$ ☒
- 2 $(0, 2)$ ☒
- 3 $(0, 5)$ ☒
- 4 None of these

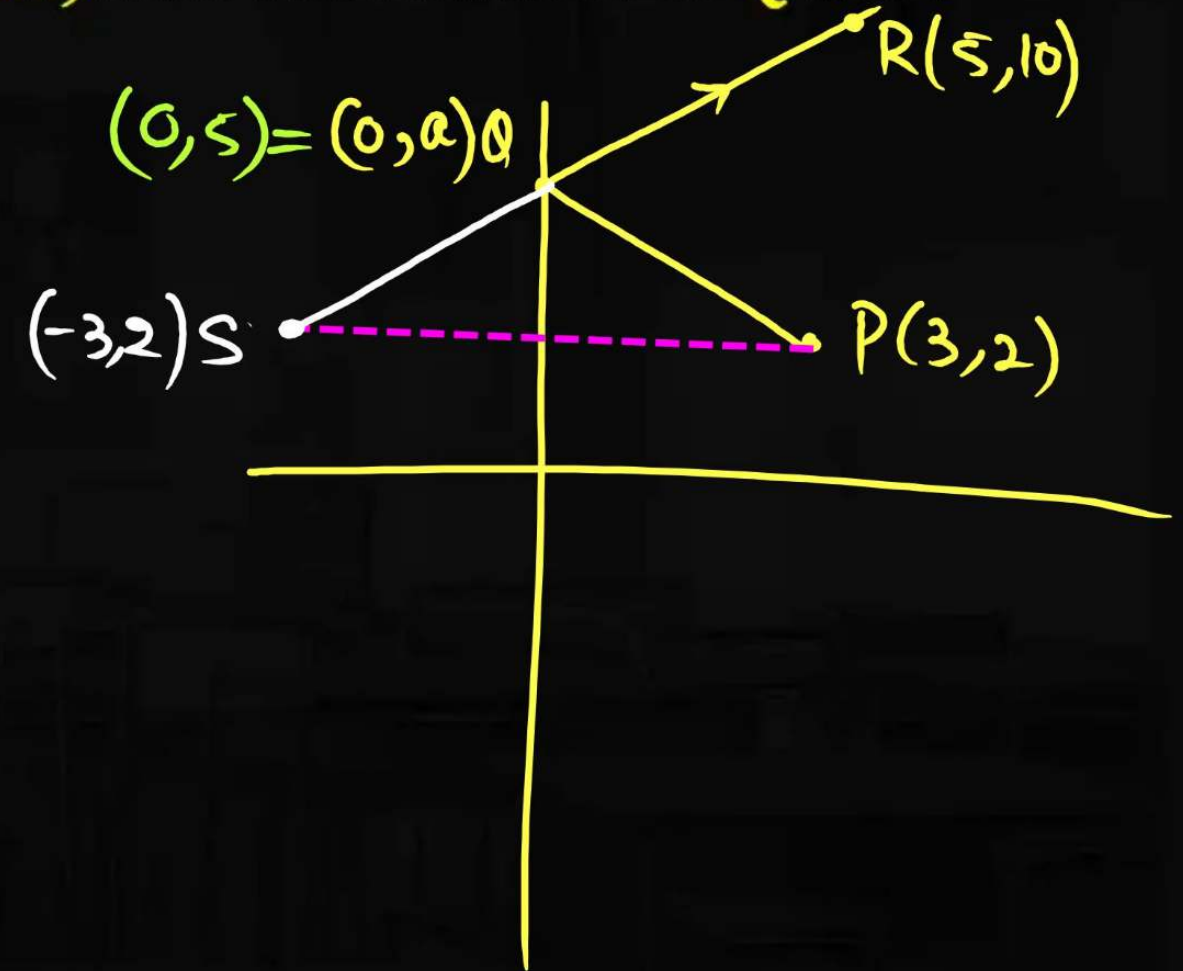
Homework.

$$\text{slope of RS} = \text{slope of QS}$$

$$\frac{10-2}{5+3} = \frac{a-2}{0+3}$$

$$\Rightarrow \frac{8}{8} = \frac{a-2}{3}$$

$$\Rightarrow a = 5$$



QUESTION

If one side of a Rhombus has end points $(4, 5)$ and $(1, 1)$, then the **maximum area** of the Rhombus is :

1 ✗ 50 sq. units

2 ✓ 25 sq. units

3 ✗ 30 sq. units

4 ✗ 20 sq. units

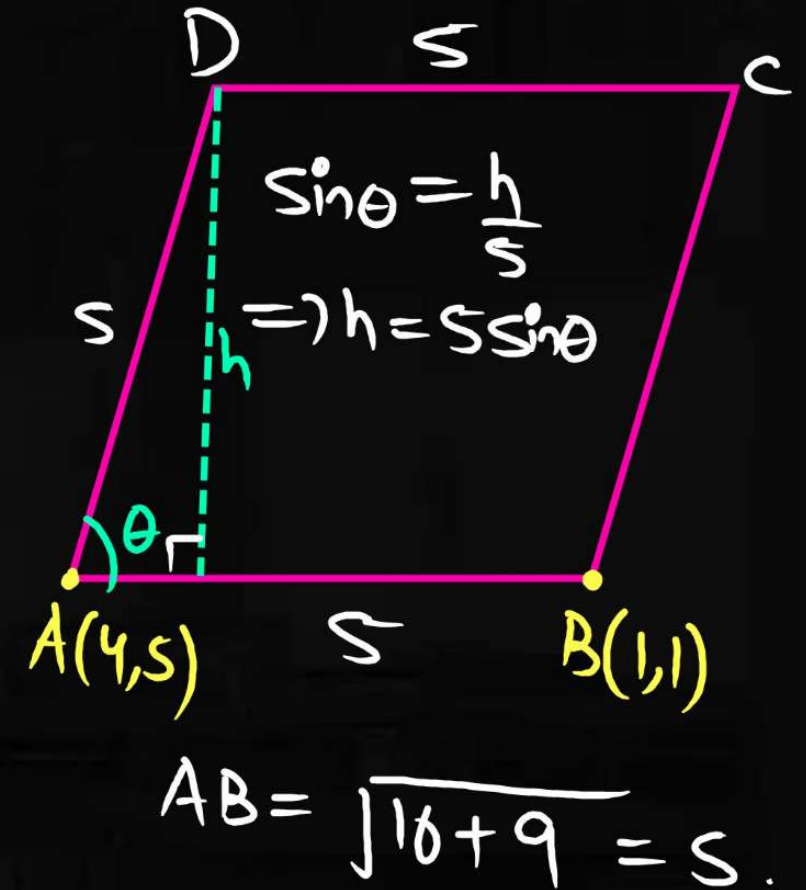
Sol)

$$\text{Area of Rhombus} = \frac{(s+s) \times h}{2}$$

$$\Rightarrow \text{Area of Rhombus} = \frac{s \times s \sin \theta}{2}$$

$$\Rightarrow \text{Area of Rhombus} = s^2 \sin \theta$$

$$\text{So Maximum Area} = 25 \times 1 = 25 \text{ Sq. units}$$





Thank
You